Bridge Layer Integration Audit

Identity Entropy Output · Formation Context · Coalescence Tracking · Parser / Runtime / Export

Doc ID	AUD-2026-01	Audit phase	________________
Ref: bridge	vsim_bridge_doc.tex	Ref: CR	vsim_code_request.tex · CR-2026-01
Target build	v5.2.0-internal	Auditor	________________
Date opened	2026-05-31	Date closed	________________

Source of truth: `include/vsim/vsim_document.hpp` Parser: `src/vsim/vsim_parser.cpp` Runtime: `include/vsim/vsim_runtime.hpp` Reference: `VSIM_REFERENCE.md`

Contents

§ 1 Audit Scope

This audit tracks integration of the bridge layer — new parser sections, runtime wiring, export artifacts, observer integration, and viewer overlay — against the stable VSIM language as of **v5.0.2 / VSIM_REFERENCE.md**. Status cells and developer notes are left blank for the implementing team to complete as tasks land. Tick checkboxes when verified against a real run.

Legend: NEW = does not exist in current reference PENDING = task open PASS = verified FAIL = regression or blocked N/A = not applicable TBD = decision pending

§ 2 Parser Coverage Audit

Cross-reference against `vsim_parser.cpp` and `vsim_document.hpp`. New bridge sections are marked NEW. All existing sections should already parse; verify they are not broken by the bridge changes.

Block / Section	Struct	Pre-bridge	Post-bridge	☐	Dev / PR	Notes
`[project]`	`ProjectSection`	PASS	____________	☐	—	no change expected
`[seed]`	`SeedSection`	PASS	____________	☐	—	no change expected
`[material]`	`MaterialSection`	PASS	____________	☐	—	no change expected
`[run]`	`RunSection`	PASS	____________	☐	—	no change expected
`[environment]`	`EnvironmentSectio`	PASS	____________	☐	—	no change expected
`[export]`	`ExportSection`	PASS	____________	☐	—	DEC-01 may rename new sub-key
`[observe]`	`ObserveSection`	PASS	____________	☐	—	add `"identity_entro` metric
`[variance]`	`VarianceSection`	PASS	____________	☐	—	no change expected
`[kernel]`	`raw_sections`	PASS	____________	☐	—	no change expected
`[while]`	`WhileSection`	PASS	____________	☐	—	no change expected
`[loop]`	`LoopSection`	PASS	____________	☐	—	no change expected
`[until]`	`UntilSection`	PASS	____________	☐	—	no change expected

Block / Section	Struct	Pre-bridge	Post-bridge	☐	Dev / PR	Notes
`[analysis.sampling]`	`VsimSamplingSecti`	PASS	______	☐	—	no change expected
`[analysis.scale_sampling]`	`VsimScaleSampling` `on`	PASS	______	☐	—	no change expected
`[verify.*]`	`VerifySection`	PASS	______	☐	—	no change expected
`[formation]`	`FormationContex` NEW	N/A	______	☐	—	CR-01 · new parser module
`[formation.initia`	`FormationContex` NEW	N/A	______	☐	—	CR-01 · sub-table of `[formation]`
`[outputs.identity_`	`IEOutputSection` NEW	N/A	______	☐	—	CR-03 · DEC-01 bracket TBD

§ 3 Runtime Wire Audit

Cross-reference against `VsimRuntime` in `vsim_runtime.hpp`. Existing runtime functions are listed to confirm they remain undisturbed. Bridge functions are new and must be added.

Function / path	CR task	Pre-bridge	Post-bridge	☐	IC?	Notes
`pace_step()`	—	WIRED	______	☐	—	no change expected
`pace_resim()`	—	WIRED	______	☐	—	no change expected
`run_chemistry_pass()`	—	WIRED	______	☐	—	coalescence gate hooks here (CR-06)
`eval_variance()`	—	WIRED	______	☐	—	no change expected
`eval_n_evolution()`	—	WIRED	______	☐	—	no change expected
`run_while_guards()`	—	WIRED	______	☐	—	no change expected
`run_batch()`	—	WIRED	______	☐	—	no change expected
`eval_observe_metrics()`	CR-08	WIRED	______	☐	—	add `"identity_entropy"` case
`load_formation_contex`	CR-01	N/A	______	☐	IC-04	must not serialise to `.dynx`

Function / path	CR task	Pre-bridge	Post-bridge	☐	IC?	Notes
`validate_formation()`	CR-02	N/A	________	☐	—	warning only — must not block run
`init_ie_writer()`	CR-03, 04	N/A	________	☐	IC-06	no-op when disabled
`populate_ie_record()`	CR-05	N/A	________	☐	IC-01	must not mutate particle state
`coalescence_gate()` hook	CR-06	N/A	________	☐	IC-02	read-only on fire
`compute_formation_mem`	CR-07	N/A	________	☐	—	wires into `IERecord.formation_me`
`register_ie_viewer_ov`	CR-10	N/A	________	☐	IC-03	viewer reads `.ie`; never writes back

§4 Output Artifact Inventory

Verify each artifact is present after a successful surface demo run (`identity_entropy_surface_demo`, from bridge doc §9).

Artifact	Type	Expected path	Export flag	Exists	Non-empty	Viewer-safe	Notes
positions	`.xyz`	`output/run.xyz`	`write_xyz`	☐	☐	PASS	ground truth; must not change
trajectory	`.xyzf`	`output/run.xyzf`	`write_xyzf`	☐	☐	PASS	ground truth; must not change
analysis JSON	`.json`	`output/analysis.json`	`write_analysis_json`	☐	☐	PASS	no change expected
events JSONL	`.jsonl`	`output/events.jsonl`	`write_events_json`	☐	☐	PASS	coalescence events should appear here
symbolic trace	`.json`	`output/symbolic_trace.json`	`write_symbolic_trace_json`	☐	☐	PASS	no change expected
report	`.md`	`output/report.md`	`write_report_md`	☐	☐	PASS	no change expected
manifest	`.json`	`output/manifest.json`	`write_manifest_json`	☐	☐	PASS	should register `.ie` in registry
identity entropy sidecar NEW	`.ie`	`output/run.ie`	`outputs.iden`	☐	☐	________	IC-01: derived only

> **Viewer contract:** The `.ie` file is an overlay sidecar, not a particle-state file. It must satisfy `must_not_mutate_state = true` per the viewer data contract (`view_contract.hpp`). Confirm this with the viewer team before CR-10 lands.

§ 5 .ie File Format Spot-Check

After a demo run, open the `.ie` file and verify each structural property. Fill in the "Observed" column from the actual file.

Property	Expected	Observed	☐	Notes
Header row present	`frame identity_id J dataloss ...`	____________	☐	
Column count	15	____________	☐	frame + id + 12 vars + flags
`J` range	$[0.0, 1.0]$ for all rows	____________	☐	IC-07: clamp enforced in kernel
`dataloss` invariant	`dataloss = 1 - J`	____________	☐	enforced by `IERecord::sync()`
`recoverability` inv.	`recoverability = 1 - projection_loss`	____________	☐	enforced by `IERecord::sync()`
`formation_memory` f0	1.0 (fully declared context)	____________	☐	AC-04
`coalescence_loss`	> 0.0 on event frame	____________	☐	AC-06 · requires DEC-05 resolved
SUMMARY block present	`# SUMMARY / # rows / # mean_J ...`	____________	☐	AC-02
File precision	6 decimal places	____________	☐	controlled by `precision` key
Row count matches frames	rows = max_steps / write cadence	____________	☐	`write_per_frame = true`

§ 6 Existing Script Compatibility Matrix

Run each existing script against the bridge build and verify no regressions. Fill each cell with ✓ (pass), × (fail), or N/A. Parser errors, runtime crashes, and missing output files all count as failures.

Script	Parses	Runs to end	Outputs OK	Verify pass	No new warnings	Run by	Date
`suite_metal_steam_properties.vsim`							

Script	Parses	Runs to end	Outputs OK	Verify pass	No new warnings	Run by	Date
suite_gas_plasma_properties.vsim							
demo_day72_nuclear_clad.vsim							
[add scripts here]							
[add scripts here]							
identity_entropy_surface_demo							

Regression priority: The three reference scripts (`suite_metal_steam`, `suite_gas_plasma`, `nuclear_clad`) must all pass before any bridge task is marked complete. These scripts have no `[formation]` block — the parser must handle their absence gracefully without emitting spurious warnings on non-applicable material types.

§7 Verification Run Log

Record each verification run as tasks land. One row per run.

Run ID	Script	Build / branch	AC tested	Result	Output dir	Run by	Date
—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—

§8 Open Issues Register

Pre-populated from known open decisions (CR-2026-01). Add new issues as they surface.

`ISS-01` Schema bracket for identity entropy output config TBD

`[outputs.identity_entropy]` vs `[export.identity_entropy]`. Must be resolved before CR-03 can land. Affects all downstream script examples and the bridge doc §9 demo script.

Ref: DEC-01 · CR-03 Owner: ________ Due: ________ Resolved: ________

`ISS-02` Formula for `dataentropy`, `hidden_activity`, `projection_loss` TBD

Three of the twelve IE variables have no defined computation formula. Until resolved, CR-05 emits `0.0` with flag bit `0x1`. Requires physics model input. Blocks full variable set completion.

Ref: DEC-02 · CR-05 Owner: ________ Due: ________ Resolved: ________

`ISS-03` `lineage_depth` tracking strategy TBD

Per-object counter vs global lineage registry. Choice affects memory usage in large runs and accuracy for re-coalesced objects.

Ref: DEC-03 · CR-05 Owner: ________ Due: ________ Resolved: ________

`ISS-04` Projection operator A_n for coalescence residual TBD

`coalescence_residual()` takes caller-supplied `projected_identity` vector. Who computes the projection, and how? Bridge doc defines $R_{\text{coal}}^{(n)} = C_n - A_n(I^\star_{(n+1)})$ but A_n is not yet specified. Blocks CR-06 and AC-06.

Ref: DEC-05 · CR-06 Owner: ________ Due: ________ Resolved: ________

`ISS-05` ______________________________ PENDING

[fill in as issues surface]

Ref: ________ Owner: ________ Due: ________ Resolved: ________

`ISS-06` ______________________________ PENDING

[fill in as issues surface]

Ref: ________ Owner: ________ Due: ________ Resolved: ________

§9 Sign-off

This audit is considered closed when all P0 and P1 acceptance criteria in CR-2026-01 are checked, the compatibility matrix shows no regressions, and the four pre-populated open issues are resolved or formally deferred. P2 items (CR-10, AC-10) may be deferred to a follow-on milestone with documented rationale.

Implementing developer	*Date*
Reviewer	*Date*
Integration lead	*Date*
Final audit status	*Build merged*

Next step (step 3): Scripting examples document — canonical `.vsim` scripts demonstrating the bridge integration at minimum, standard, and maximum instrumentation levels. Should include: surface deposition (bridge doc §9 demo), powder bed with coalescence tracking, batch sweep with `formation` axis, and a nuclear cladding variant adding identity entropy alongside the existing Day-72 script.